

Dynamical Modelling of Bleeding-Ischemia Risk Surfaces for Antithrombotic De-escalation after PCI Based on Hemoglobin, Renal Function, and Prior Bleeding

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Abstract

Antithrombotic therapy after percutaneous coronary intervention (PCI) involves a time-dependent trade-off between ischemic protection and bleeding liability. Established risk frameworks identify clinically relevant bleeding and ischemic factors, but they were not designed to represent how a treatment-switching boundary may change jointly with patient vulnerability and elapsed time after PCI. We developed a theoretical dynamical proof-of-concept to examine this problem using hemoglobin, renal function, prior bleeding, and time from a 30-day post-acute coronary syndrome landmark. The framework was literature-informed but not fitted to patient-level outcomes. Hemoglobin and estimated glomerular filtration rate were represented by continuous vulnerability functions centered on clinically recognized risk regions, while prior bleeding was represented as a binary state. Time-dependent bleeding and ischemic hazards were coupled to normalized antithrombotic treatment intensity and propagated through latent burden states. A prespecified loss function combined bleeding and ischemic liabilities and was evaluated over candidate treatment-switching times to identify a theoretical loss-minimizing switching time, conditional on the specified parameterization. Structural uncertainty was examined using Latin-hypercube sampling and partial rank correlation analysis, while stochastic perturbations were used to assess variation in the simulated clinical states. Under the reference parameterization, the loss-minimizing switching time occurred approximately 3.0 months after the 30-day landmark for the reference profile (hemoglobin 11.5 g/dL, eGFR 45 mL/min/1.73 m², no prior bleeding). Earlier switching emerged under greater bleeding vulnerability: the theoretical minimum occurred at the 30-day boundary for the combined high-bleeding-risk scenario, 0.6 months after the landmark with prior bleeding, and 2.4 months with isolated anemia, compared with 4.2 months for the renal-vulnerability scenario and 4.6 months for the low-vulnerability profile. Across 600 illustrative stochastic perturbations, the median loss-minimizing switching time was 2.8 months after the landmark, with a 2.5th–97.5th simulation percentile range of 0.0–4.0 months; 14.7% of simulations selected the earliest permitted boundary. Global sensitivity analysis showed that switching time was most responsive to the assumed treatment-related bleeding effect, baseline ischemic and bleeding hazard scales, and the relative weighting of bleeding and ischemic losses. Numerical solutions were stable under alternative integration settings. These findings demonstrate the mathematical behavior of the proposed framework rather than the clinical effectiveness or optimal timing of antithrombotic de-escalation. The analysis shows how clinically recognized vulnerability factors, elapsed post-PCI time, treatment effects, and explicit loss assumptions can be represented within a time-dependent switching problem. The resulting switching times are conditional simulation outputs and should not be interpreted as treatment recommendations. Patient-level parameter estimation, clinical calibration, and external validation are required before the framework can be evaluated for predictive or decision-support use.

Keywords: Antithrombotic De-escalation, Bleeding Ischemia Trade-off, Dynamical Modelling, Percutaneous Coronary Intervention, Switching-Time Analysis

1. Background and Aim

Antithrombotic management after percutaneous coronary intervention (PCI) requires balancing protection from ischemic events against the risk of clinically relevant bleeding. This balance changes over time and differs according to patient characteristics and treatment exposure. Contemporary European guidelines recognize this temporal dimension by allowing antiplatelet de-escalation beyond the first 30 days after acute coronary syndrome (ACS) in selected patients, whereas de-escalation during the first 30 days is generally not recommended (ESC, 2023).

Several established risk models identify clinical factors associated with bleeding after PCI. The Academic Research Consortium for High Bleeding Risk (ARC-HBR) classifies hemoglobin <11 g/dL

and severe chronic kidney disease with eGFR <30 mL/min/1.73 m² as major high-bleeding-risk criteria, while moderate renal impairment and milder anemia contribute additional bleeding-risk information (Urban et al., 2019). Previous spontaneous bleeding is also incorporated into established bleeding-risk definitions (Urban et al., 2019 and Costa et al., 2017). Similarly, PRECISE-DAPT includes hemoglobin, renal function, and previous bleeding among variables used to estimate bleeding risk during antiplatelet treatment (Costa et al., 2017 and Gragnano et al., 2022).

These variables were selected for the present framework because they provide a parsimonious set of clinically interpretable inputs rather than because they constitute a complete post-PCI risk model. Renal dysfunction is particularly relevant to the competing-risk problem because it has been associated with both coronary thrombotic events and major bleeding in the PARIS framework, whereas anemia has been associated predominantly with bleeding risk (Baber et al., 2016). Previous studies have also reported associations between anemia and adverse outcomes after PCI. Hemoglobin, renal function, and previous bleeding therefore provide clinically grounded variables with which to examine the mathematical behavior of a bleeding ischemia trade-off, while other established prognostic factors are intentionally excluded from the present proof-of-concept.

Randomized trials provide further evidence that the balance between ischemic protection and bleeding can change when antiplatelet therapy is modified after PCI. TOPIC examined switching from potent P2Y₁₂ inhibition to clopidogrel after ACS (Cuisset et al., 2017), TROPICAL-ACS evaluated guided de-escalation (Sibbing et al., 2017), and TALOS-AMI evaluated unguided de-escalation after stabilization following acute myocardial infarction (Kim et al., 2021). Other trials, including TWILIGHT (Mehran et al., 2019), SMART-CHOICE (Hahn et al., 2019), STOPDAPT-2 (Watanabe et al., 2019), HOST-REDUCE-POLYTECH-ACS (Kim et al., 2020), and MASTER DAPT (Valgimigli et al., 2021), evaluated different strategies for reducing or modifying antiplatelet exposure. These studies differ in patient populations, treatment strategies, timing, and endpoint definitions and therefore should not be interpreted as estimating a common de-escalation effect. Instead, they establish the broader clinical rationale for considering treatment intensity as a time-dependent component of the post-PCI bleeding ischemia balance.

Existing clinical scores and consensus criteria are valuable for risk stratification, but they were not designed to represent a continuously evolving treatment-switching boundary defined jointly by clinical vulnerability, elapsed post-PCI time, and explicit assumptions about the relative consequences of bleeding and ischemic events. This leaves a methodological question: whether the post-PCI bleeding ischemia trade-off can be represented as a dynamical system in which a treatment-switching boundary changes as patient vulnerability and ischemic pressure evolve over time.

The present study therefore develops a theoretical mathematical proof-of-concept using hemoglobin, renal function, previous bleeding, and elapsed time after a 30-day post-ACS landmark to examine a time-dependent antithrombotic switching problem. Published evidence is used to define clinically interpretable variables, guideline-supported landmarks, and the expected directions of selected model relationships. Parameters that are not directly identifiable from the literature, including baseline hazard scales, treatment-effect coefficients, temporal decay parameters, and loss weights are treated as theoretical structural assumptions rather than clinical estimates. The resulting switching times and risk quantities are consequently conditional simulation outputs, not estimates of optimal treatment timing or individual patient risk. The purpose of the framework is to establish a mathematically explicit and testable structure that can subsequently be estimated and evaluated using patient-level PCI data.

2. Methods

2.1 Study Design and Modelling Framework

This study developed a literature-informed theoretical mathematical framework to examine the time-dependent balance between bleeding and ischemic liabilities after percutaneous coronary intervention (PCI). The analysis was designed as a mathematical proof-of-concept rather than a fitted clinical prediction model or treatment-recommendation tool. No patient-level outcomes were used to estimate model coefficients. Published evidence served three specific purposes: to identify clinically interpretable variables, to define the 30-day post-acute coronary syndrome treatment landmark, and to establish the expected directions of selected relationships within the model. Numerical quantities that were not directly identifiable from published evidence were specified as reference simulation parameters and were subsequently examined through sensitivity and uncertainty analyses. These quantities were not interpreted as empirically estimated clinical effects.

The model used three patient-level characteristics: hemoglobin (Hb), renal function represented by estimated glomerular filtration rate (eGFR), and previous spontaneous bleeding. These variables were selected as a parsimonious set of clinically interpretable inputs for examining the theoretical bleeding–ischemia trade-off, rather than as a complete set of predictors of post-PCI outcomes. Their clinical relevance was supported by established bleeding-risk frameworks, including ARC-HBR and PRECISE-DAPT (Urban et al., 2019 and Costa et al., 2017). Evidence from subsequent PRECISE-DAPT evaluation and the PARIS risk framework further supported their relevance to bleeding risk and, particularly for renal dysfunction, overlapping thrombotic risk (Gagnano et al., 2022 and Baber et al., 2016).

The model distinguished explicitly between literature-supported clinical anchors and theoretical structural assumptions. Clinical anchors informed variable selection, clinically recognized risk regions, and the treatment landmark. In contrast, baseline hazard scales, treatment-effect coefficients, temporal decay parameters, latent-state coefficients, relative loss weights, and switching costs were treated as theoretical quantities used to examine model behavior. The resulting trajectories, cumulative probabilities, and switching times were therefore conditional simulation outputs rather than estimates of individual patient risk or clinically optimal treatment timing.

2.2 Observable Variables

Three observable clinical variables were included in the proof-of-concept framework: hemoglobin (Hb), estimated glomerular filtration rate (eGFR), and previous spontaneous bleeding. These variables were represented as model inputs rather than fitted predictors, and their inclusion was intended to provide a parsimonious clinical basis for examining the theoretical bleeding–ischemia trade-off. Hemoglobin was represented as a continuous variable in g/dL. The ARC-HBR definition was used only as a clinical reference for constructing the hemoglobin vulnerability function, with Hb <11 g/dL recognized as a major high-bleeding-risk criterion (Urban et al., 2019). This threshold was not imposed as a discrete switching rule in the model. Instead, hemoglobin entered the dynamical system through a continuous transformation, allowing bleeding vulnerability to vary gradually over the simulated hemoglobin range. Previous studies have also reported associations between anemia and adverse outcomes after PCI; these observations supported the clinical relevance and expected direction of the hemoglobin component but were not used to estimate its mathematical effect size.

Renal function was represented continuously by eGFR in mL/min/1.73 m². ARC-HBR identifies eGFR <30 mL/min/1.73 m² as a major high-bleeding-risk criterion and eGFR 30–59 mL/min/1.73 m² as a minor criterion (Urban et al., 2019). These categories were used as clinical reference regions rather than discrete

mathematical cutoffs. Renal function entered the model through a continuous vulnerability transformation. Its inclusion as a potentially overlapping bleeding–ischemia factor was additionally motivated by the PARIS framework, in which renal impairment was informative for both coronary thrombotic and major bleeding events (Baber et al., 2016). The magnitude of the renal contribution in the present model was specified theoretically and was not estimated from PARIS or ARC-HBR.

Previous spontaneous bleeding was represented as a binary state, $P \in \{0,1\}$, where $P = 1$ indicated a history of previous spontaneous bleeding and $P = 0$ indicated its absence. Its inclusion was supported by ARC-HBR and PRECISE-DAPT, both of which recognize previous bleeding as clinically relevant to subsequent bleeding risk (Urban et al., 2019 and Costa et al., 2017). Binary encoding was adopted as a deliberate simplification for the present proof-of-concept and does not represent differences in bleeding severity, anatomical site, frequency, or time elapsed since the previous event. The effect assigned to P within the model was therefore treated as a theoretical structural parameter rather than an effect estimate derived from these clinical risk models.

2.3 Model Formulation

$$\text{Hemoglobin vulnerability } q_{Hb} = \left[1 + \exp \left\{ \frac{Hb - 11.5}{0.85} \right\} \right]^{-1}$$

$$\text{Renal vulnerability } q_G = \left[1 + \exp \left\{ \frac{eGFR - 50}{14} \right\} \right]^{-1}$$

$$\text{Prior-bleeding state } P \in \{0,1\}$$

$$\text{Treatment intensity } u(t) = 1 \text{ for potent DAPT; } u(t) = 0.55 \text{ for the illustrative de-escalated state}$$

$$\text{Bleeding hazard } h_{B(t)} = b_0 \exp\{\beta_H q_{Hb} + \beta_G q_G + \beta_P P + \eta_B u(t)\}$$

$$\text{Ischemic hazard } h_{I(t)} = i_0 \exp\{\gamma_H q_{Hb} + \gamma_G q_G + \gamma_P P - \eta_I u(t) - \delta_I t\}$$

$$\text{Latent bleeding burden } \frac{dB}{dt} = k_B h_B(t)[1 - B(t)] - r_B B(t)$$

$$\text{Latent ischemic burden } \frac{dI}{dt} = k_I h_I(t)[1 - I(t)] - r_I I(t)$$

$$\text{Cumulative bleeding probability } P_{B(T)} = 1 - \exp \left\{ - \int_0^T h_B(s) ds \right\}$$

$$\text{Cumulative ischemic probability } P_{I(T)} = 1 - \exp \left\{ - \int_0^T h_I(s) ds \right\}$$

$$\text{Expected clinical loss } L(\tau) = \lambda_B P_B(T|\tau) + \lambda_I P_I(T|\tau) + c_s 1(\tau < T)$$

$$\text{Loss-minimizing switching time } \tau^* = \text{set of 56 candidate switching times}$$

$$\text{Global sensitivity } \text{PRCC}(\theta_j, \tau^*) = \text{Corr}[\text{rank-residual}(\theta_j), \text{rank-residual}(\tau^*)]$$

The hemoglobin and renal vulnerability functions provide continuous representations of clinical vulnerability rather than binary implementations of ARC-HBR thresholds. Their centers and scales define the shape of the theoretical vulnerability functions and are not estimated clinical cutoffs. Prior bleeding is represented by the binary state P . Treatment intensity is normalized, with $u = 1$ representing the reference potent-treatment state and $u = 0.55$ representing an illustrative lower-exposure state; these values define a relative model contrast and should not be interpreted as pharmacological dose equivalence among individual antiplatelet agents.

The bleeding and ischemic hazards are theoretical cause-specific intensity functions under the specified parameterization. The ischemic hazard includes an explicit exponential time-decay term after the 30-day landmark, whereas the bleeding hazard remains dependent on vulnerability and treatment exposure. The

latent states $B(t)$ and $I(t)$ are dimensionless mathematical states that accumulate preceding hazard exposure while allowing recovery through the terms $r_B B(t)$ and $r_I I(t)$. They are introduced to represent path dependence and should not be interpreted as directly observed biological quantities.

For a candidate switching time τ , treatment intensity changes from the potent state to the illustrative de-escalated state at τ , and the corresponding hazards are integrated over the remaining analysis horizon to obtain model-implied cumulative bleeding and ischemic probabilities. The prespecified loss function combines these quantities using the relative weights λ_B and λ_I and includes the switching cost c_s . The reported switching time τ^* is the candidate time producing the minimum loss under the specified parameterization. It is therefore referred to as the loss-minimizing switching time rather than a clinically optimal treatment time.

2.4 Theoretical Analysis

The theoretical analysis focused on the behavior of the latent bleeding and ischemic states rather than on a clinical steady state. For finite parameter values, the ordinary differential equation system has continuous, locally Lipschitz right-hand sides over the model domain, supporting local existence and uniqueness of solutions. The normalized interval $[0,1]$ was preserved by the boundary structure of the latent states under the specified non-negative parameter assumptions. For local analysis, hemoglobin, eGFR, prior bleeding, treatment intensity, and elapsed post-PCI time were treated as fixed, allowing the B-I subsystem to be examined using the Jacobian matrix and its eigenvalues. Because elapsed time enters the ischemic hazard explicitly, the complete model is non-autonomous and a full-system equilibrium was not treated as the principal theoretical endpoint. Instead, the zero-level set of the expected-loss difference defined a time-dependent switching boundary in hemoglobin eGFR space conditional on prior bleeding, elapsed time, and the specified parameterization, this boundary represents a theoretical model property rather than an empirically estimated clinical treatment threshold.

2.5 Computational Analysis

Reference latent-state trajectories were numerically integrated using adaptive RK45 with relative and absolute tolerances of 10^{-9} and 10^{-11} , respectively, and numerical stability was assessed against DOP853 using maximum step sizes from 1.0 to 0.1 month. The theoretical switching surface was evaluated over hemoglobin values of 9.5–14.5 g/dL in 0.25-g/dL increments and eGFR values of 20–100 mL/min/1.73 m² in 5-unit increments, separately for the presence and absence of prior bleeding. At each grid point, 56 candidate switching times were evaluated over the 11-month post-landmark horizon, and the time minimizing the prespecified loss function was retained as the model-implied loss-minimizing switching time. Five illustrative scenarios low vulnerability, renal vulnerability, anemia, prior bleeding, and combined high-bleeding-risk vulnerability were additionally evaluated to examine model behavior under contrasting clinical profiles.

2.6 Robustness, Uncertainty, and Sensitivity

Structural uncertainty was examined using 320 Latin-hypercube samples over prespecified ranges for baseline hazard scales, vulnerability coefficients, treatment-effect parameters, ischemic-decay parameters, and relative loss weights. Partial rank correlation coefficients (PRCCs) were used to quantify associations between individual parameters and the model-implied loss-minimizing switching time after adjustment for the remaining sampled parameters. Local sensitivity was additionally assessed using +1% perturbations around the reference parameter values. Separately, 600 illustrative state perturbations varied hemoglobin and eGFR around the reference state and sampled prior-bleeding status to distinguish variation in observable model inputs from uncertainty in structural parameters. The resulting distributions represent simulation-based scenario dispersion rather than confidence intervals, measurement error, or empirically estimated patient-level variability.

2.8 Planned Clinical Calibration and Statistical Validation

Future clinical evaluation would require patient-level PCI data containing the index presentation, antiplatelet regimen and treatment changes, serial hemoglobin and creatinine/eGFR measurements, prior bleeding, and prospectively defined bleeding and ischemic outcomes. Structural parameters could then be estimated using longitudinal time-to-event or state-space approaches and compared with appropriate competing-risk or multi-state survival models. Model performance should be evaluated only after empirical fitting, using measures such as out-of-sample log likelihood, calibration intercept and slope, time-dependent Brier score, and decision-curve net benefit, with internal validation to assess optimism. Any clinical analysis would also require explicit handling of missing measurements, informative follow-up, time-dependent treatment exposure, confounding, and treatment-selection bias. Until such calibration and validation are completed, the parameters and switching times reported in the present study should be interpreted solely as theoretical simulation outputs.

3. Results

3.1 Structural Behaviour of the Model

Under the reference parameterization, the model produced the prespecified directional behavior of the bleeding–ischemia trade-off. Greater bleeding vulnerability shifted the loss-minimizing switching time earlier, whereas lower bleeding vulnerability permitted later switching. Lower hemoglobin, impaired renal function, and previous bleeding each altered the position of the theoretical switching boundary, with the largest shifts occurring when multiple vulnerability factors were present. These patterns represent consequences of the specified mathematical structure and parameterization rather than empirically estimated clinical associations.

3.2 Dynamic Switching Surface

The theoretical switching boundary varied with hemoglobin, eGFR, and prior-bleeding status under the reference parameterization. For the reference state (Hb 11.5 g/dL, eGFR 45 mL/min/1.73 m², no prior bleeding), the expected-loss function reached its minimum 3.0 months after the 30-day landmark. The loss-minimizing switching time occurred later in the low-vulnerability scenario (Hb 13.8 g/dL, eGFR 85 mL/min/1.73 m², no prior bleeding), at 4.6 months, and earlier in the isolated-anemia scenario (Hb 10.8 g/dL), at 2.4 months. Prior bleeding with otherwise preserved laboratory values shifted the minimum to 0.6 months after the landmark, whereas the combined-vulnerability scenario (Hb 10.5 g/dL, eGFR 25 mL/min/1.73 m², prior bleeding present) selected the earliest permitted switching time at the 30-day landmark. These switching times are conditional outputs of the theoretical model and should not be interpreted as empirically estimated treatment thresholds or clinical recommendations.

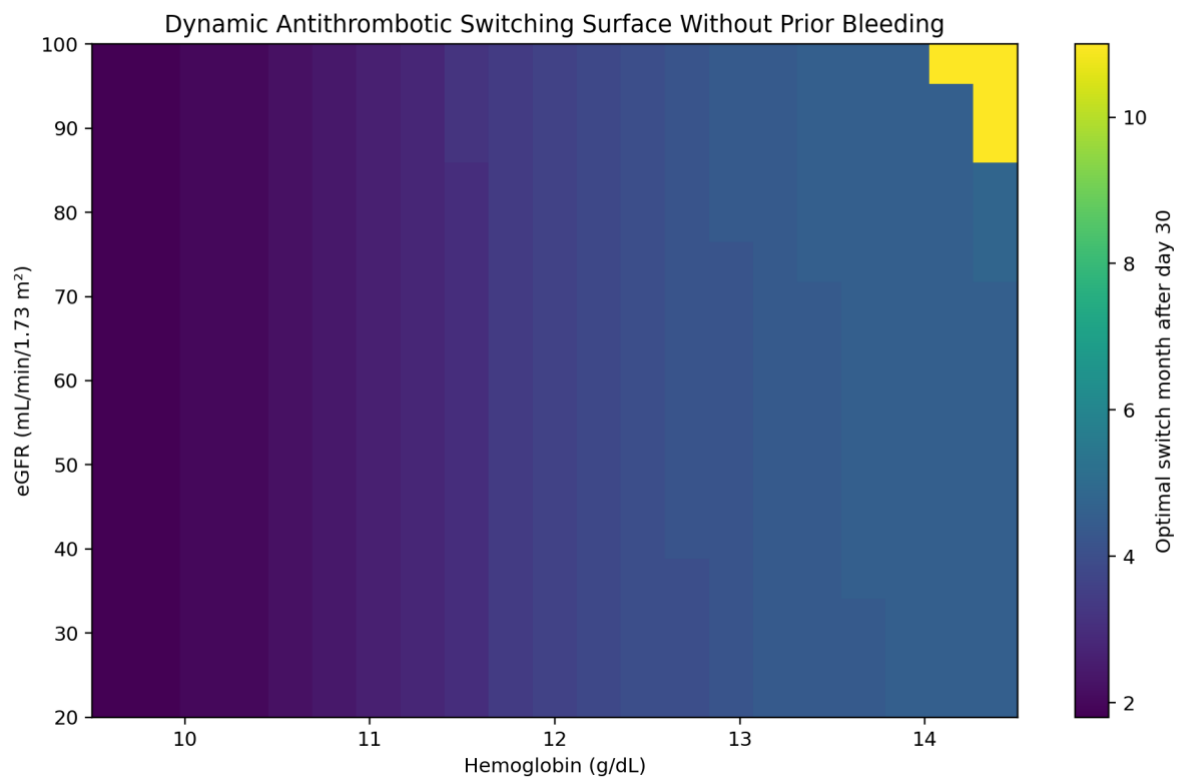


Figure 1. Hemoglobin–eGFR switching surface without prior bleeding. The color field denotes the simulation-derived loss-minimizing month after the 30-day landmark, not a clinical cutoff.

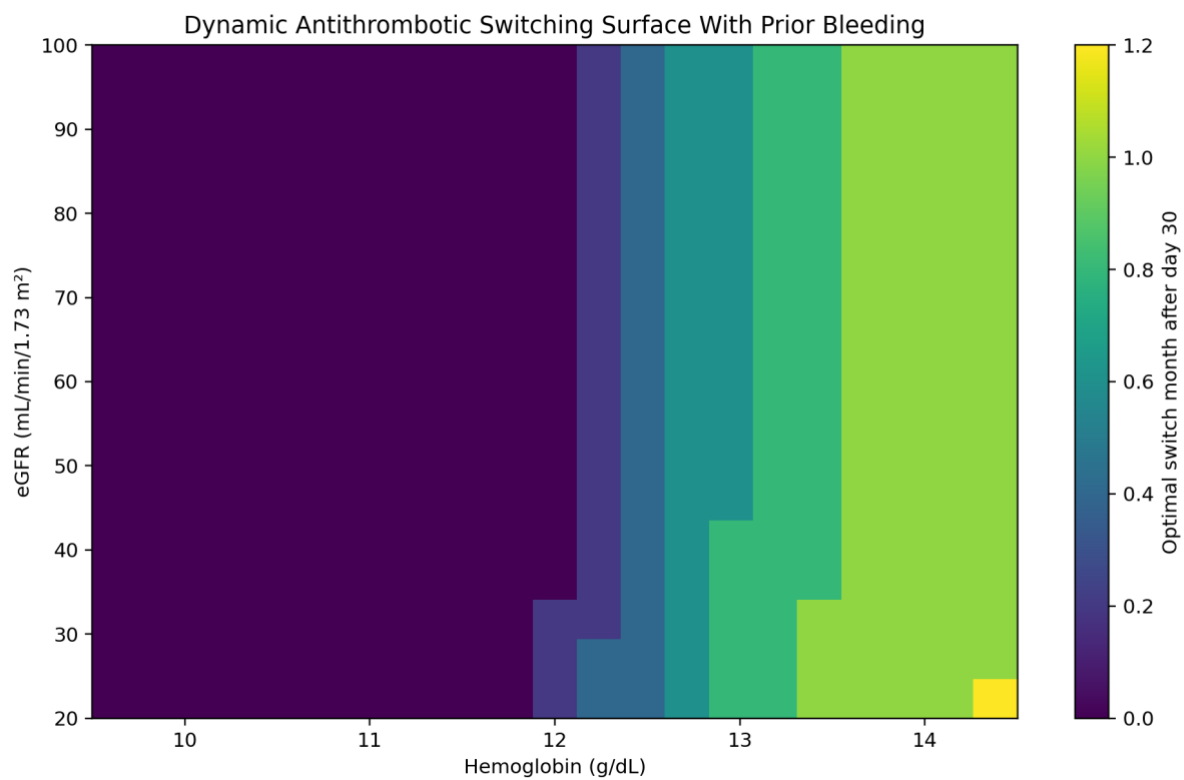


Figure 2. The same theoretical surface conditional on prior bleeding, demonstrating systematic leftward movement of the switching boundary.

3.3 Scenario Stress Tests

Profile	Hb	eGFR	Prior bleeding	Switch month	Bleed % potent	Ischemia % potent	Bleed % de-escalated	Ischemia % de-escalated
Low vulnerability	13.8	85	0	4.6000000000000005	5.2	1.2	3.82	2.01
Renal vulnerability	13.0	35	0	4.2	8.75	1.88	6.46	3.14
Anemia	10.8	75	0	2.4000000000000004	9.14	1.42	6.75	2.38
Prior bleeding	13.0	75	1	0.6000000000000001	12.52	1.41	9.3	2.36
Combined HBR	10.5	25	1	0.0	31.1	2.52	23.8	4.2

Note: All probabilities and switching times are conditional outputs of the theoretical model under the reference parameterization and specified treatment states; they are not observed event rates, calibrated patient-level risks, or clinical treatment recommendations. A switching time of 0.0 denotes selection of the earliest permitted switching boundary at the 30-day post-ACS landmark.

Across the five illustrative scenarios, the model produced the expected trade-off between lower bleeding liability and reduced ischemic protection under the de-escalated treatment state. Compared with continued potent treatment over the simulated horizon, the de-escalated state reduced model-implied bleeding probability by 1.38-7.30 percentage points while increasing model-implied ischemic probability by 0.81-1.68 percentage points. The loss-minimizing switching time occurred later in the low-vulnerability and renal-vulnerability scenarios and progressively earlier with anemia, prior bleeding, and combined vulnerability. In the combined-vulnerability scenario, the model-implied bleeding probabilities were 31.10% under the potent state and 23.80% under the de-escalated state. These comparatively high simulated values reflect the behavior of the reference model under stacked vulnerability assumptions and should not be interpreted as population event-rate estimates.

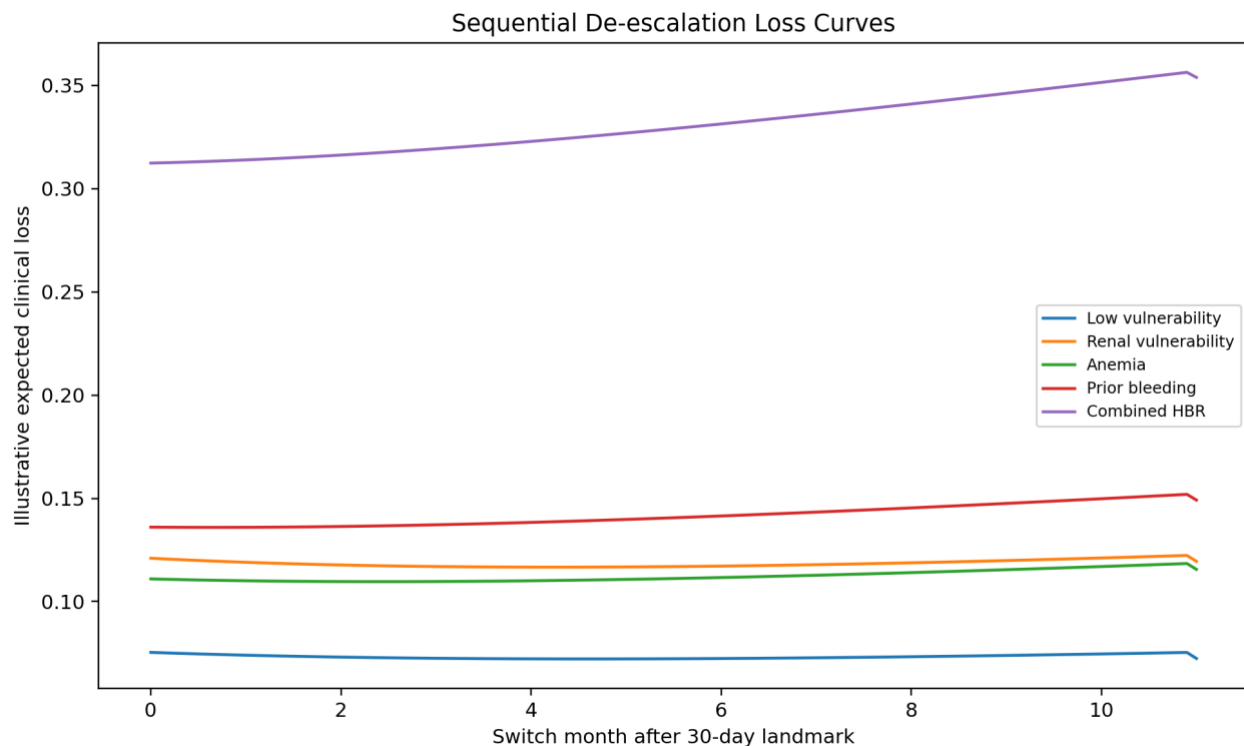


Figure 3. Sequential expected-loss curves under five vulnerability profiles. Minima move earlier as bleeding vulnerability accumulates.

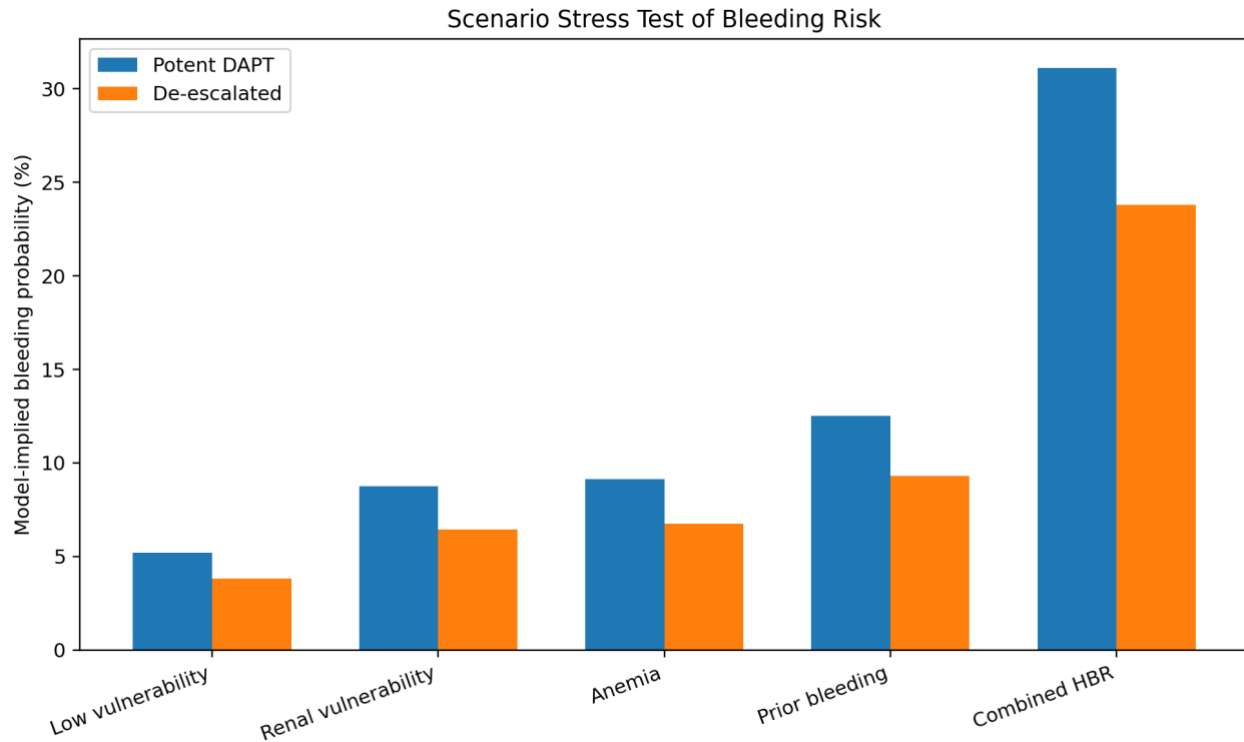


Figure 4. Scenario stress test of model-implied bleeding probability under potent and de-escalated exposure states.

3.4 Latent-State Dynamics and Regime Geometry

Under the reference parameterization, the latent ischemic state declined as the time-dependent ischemic hazard decreased, whereas the latent bleeding state remained influenced by hemoglobin, renal vulnerability, and continued treatment exposure. Consequently, the expected-loss difference evolved over time and shifted the position of the theoretical switching boundary even when the observable clinical inputs were held fixed. This behavior illustrates the path dependence generated by the model: the same hemoglobin, eGFR, and prior-bleeding state can occupy different positions relative to the switching boundary at different post-landmark times because the underlying bleeding–ischemia balance changes with elapsed time. These trajectories reflect the specified dynamical structure and should not be interpreted as empirically observed patient-state transitions.

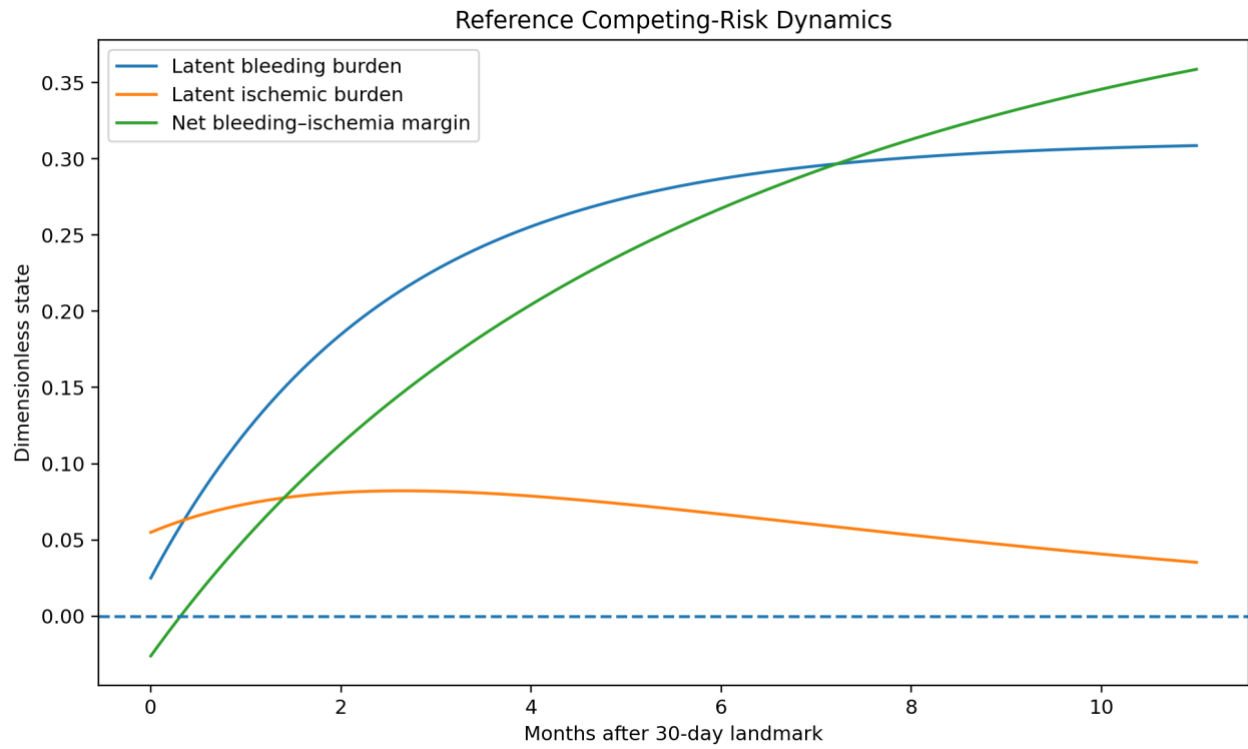


Figure 5. Reference competing-risk dynamics showing latent bleeding burden, ischemic burden, and the time-varying net margin.

3.5 Global Sensitivity and Uncertainty Quantification

Across 320 Latin-hypercube samples of the structural parameters, the loss-minimizing switching time was most strongly associated with treatment-related bleeding amplification (PRCC -0.896), baseline ischemic hazard (0.842), baseline bleeding hazard (-0.778), the bleeding loss weight (-0.730), and the ischemic loss weight (0.695). Parameters governing the renal contribution to ischemic and bleeding hazards also showed substantial associations with switching time (|PRCC| approximately 0.562 and 0.516, respectively). Across 600 illustrative state perturbations, the median loss-minimizing switching time was 2.8 months after the 30-day landmark, with a 2.5th-97.5th simulation percentile range of 0.0-4.0 months, 14.7% of simulations selected the earliest permitted switching time. These results indicate that the model-implied switching time is sensitive to both structural assumptions and variation in observable input states, rather than being represented by a single invariant cutoff.

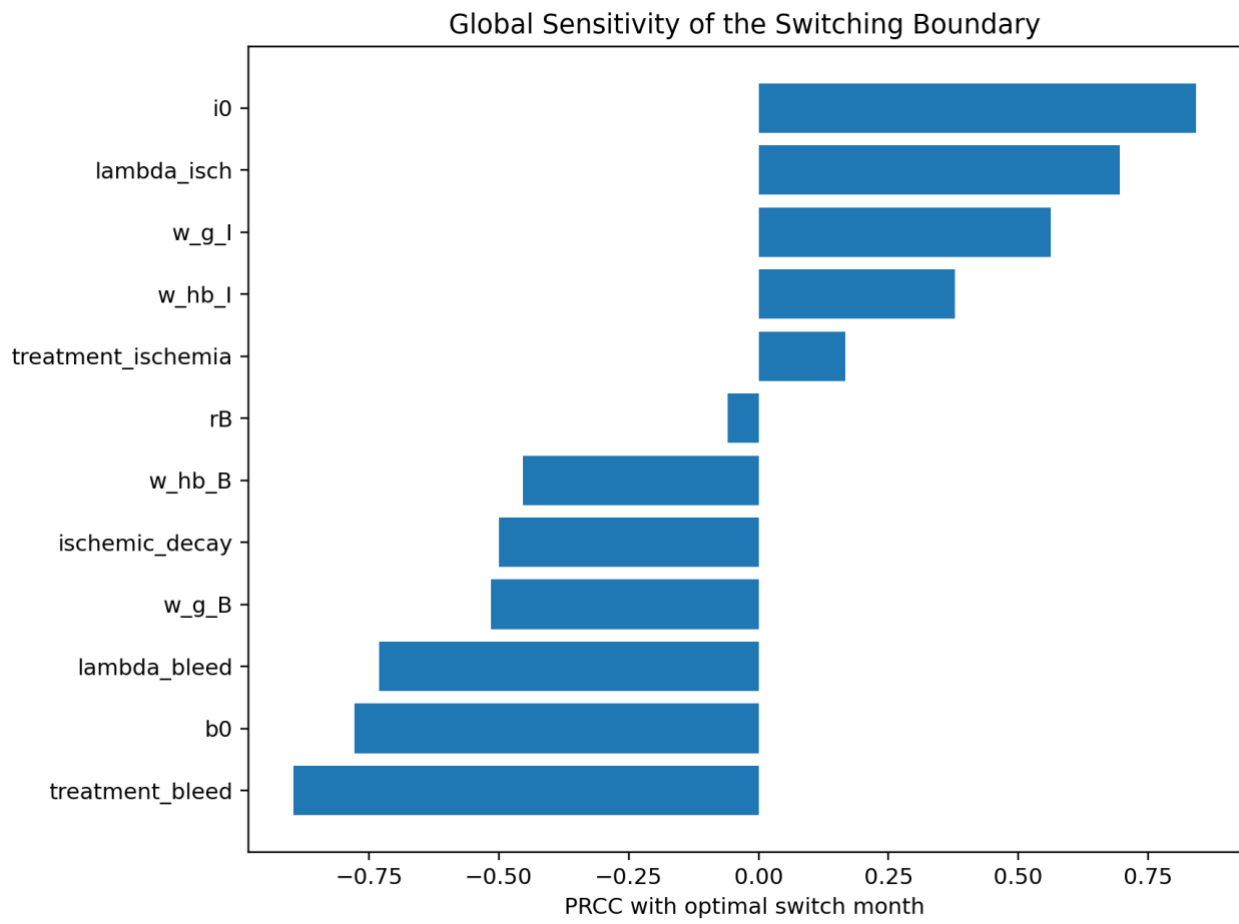


Figure 6. PRCC sensitivity of the theoretical optimal switching month to structural parameters.

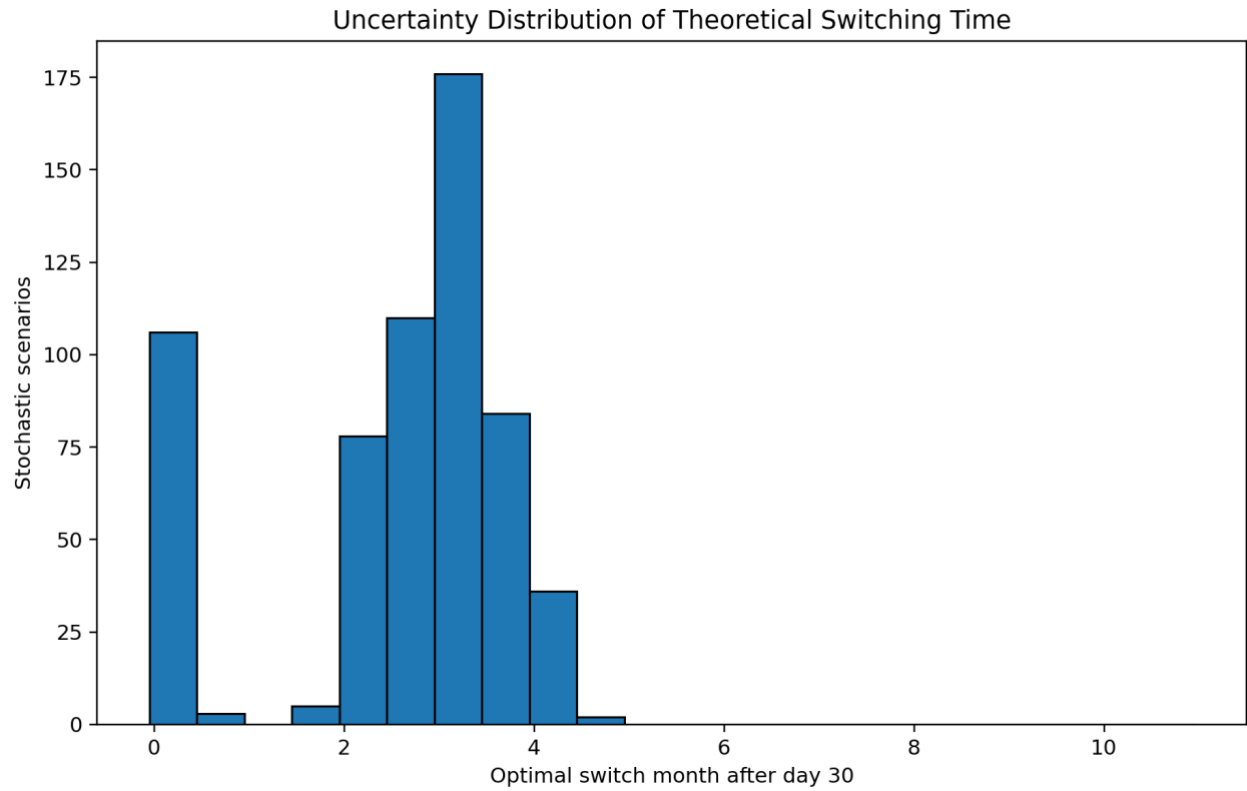


Figure 7. Distribution of theoretical switching times under stochastic perturbation of hemoglobin, eGFR, and prior-bleeding status.

3.6 Numerical Robustness

Numerical solutions were stable across the evaluated integration settings. Comparison of RK45 and DOP853 produced differences of 6.09×10^{-11} in the terminal latent bleeding state, 7.63×10^{-12} in the terminal latent ischemic state, and 7.35×10^{-11} in the terminal net-risk margin. These differences were negligible relative to the scale of the corresponding model states, indicating that the reported trajectories and switching results were not materially affected by the numerical solver configuration.

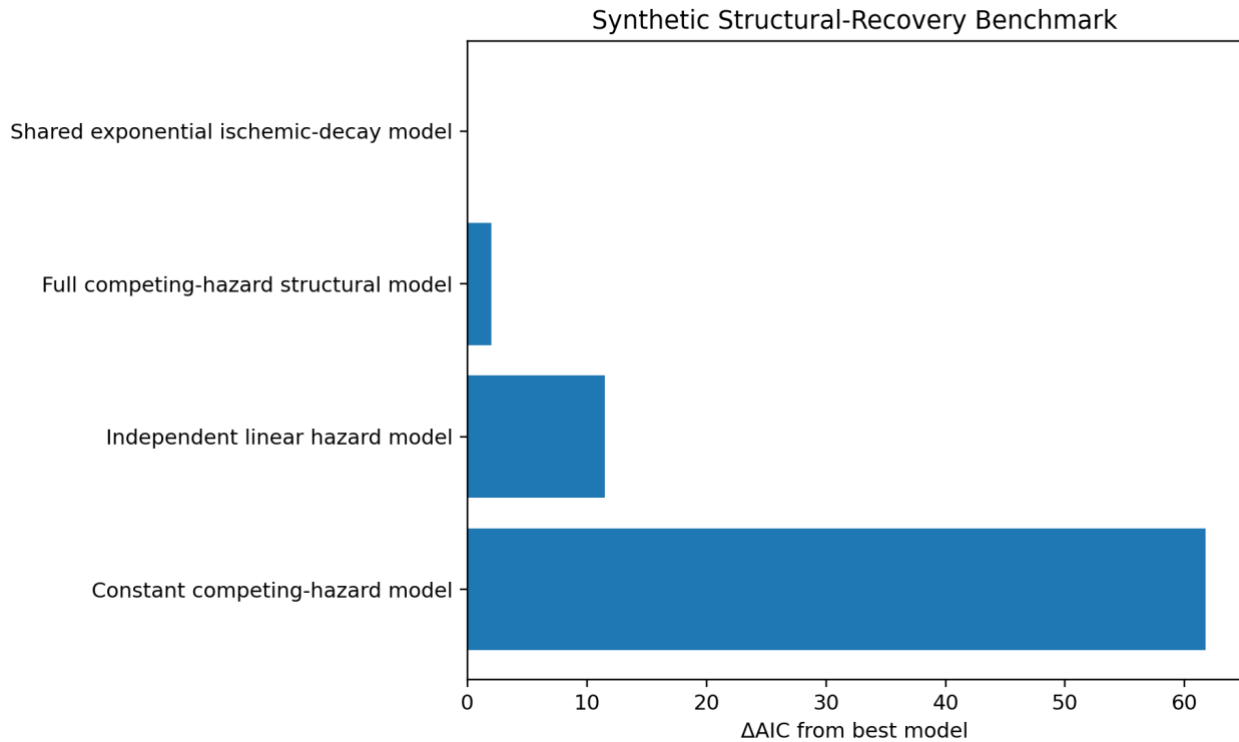


Figure 8. Synthetic AIC comparison used solely for internal structural recovery.

In the synthetic model-recovery analysis, the reduced exponential ischemic-decay model produced the lowest AIC. The full structural model was within $\Delta AIC = 2.00$, whereas the independent linear and constant-hazard alternatives yielded ΔAIC values of 11.47 and 61.75, respectively. Because the pseudo-observations were generated under prespecified theoretical assumptions, these results were interpreted only as an internal structural comparison under controlled simulation conditions and not as evidence of clinical predictive performance or superiority of the full model.

4. Discussion

This theoretical proof-of-concept shows that the post-PCI bleeding–ischemia trade-off can be represented as a time-dependent switching problem in which the position of the loss-minimizing boundary depends jointly on clinical vulnerability and elapsed time after the 30-day landmark. Under the reference parameterization, lower hemoglobin, impaired renal function, and previous bleeding shifted the model-implied switching time earlier, whereas lower-vulnerability profiles permitted later switching. The reference profile reached its minimum expected loss at 3.0 months after the landmark, compared with 4.6 months in the low-vulnerability scenario, 2.4 months with isolated anemia, 0.6 months with previous bleeding, and the earliest permitted boundary in the combined-vulnerability scenario. These findings describe the behavior of the specified mathematical system and should not be interpreted as empirically estimated treatment timings.

A central feature of the model is that switching time is not determined solely by baseline vulnerability. The latent ischemic state decreases as the explicitly time-dependent ischemic component declines, whereas the bleeding state remains influenced by hemoglobin, renal function, previous bleeding, and treatment exposure. Consequently, an unchanged clinical profile may occupy a different position relative to the theoretical switching boundary at different post-landmark times. This path dependence distinguishes the proposed formulation from a purely static representation of baseline risk, although

whether such additional temporal structure improves clinical prediction or decision utility remains to be established empirically.

The direction of the simulated trade-off is broadly consistent with the clinical rationale underlying contemporary antiplatelet de-escalation strategies. Randomized studies including TOPIC, TROPICAL-ACS, TALOS-AMI, TWILIGHT, HOST-REDUCE-POLYTECH-ACS, and MASTER DAPT have shown that modification or reduction of antiplatelet exposure can reduce bleeding in selected post-PCI populations, although the interventions, populations, timing, and endpoint definitions differ substantially among trials. The present model does not reproduce or validate these trial effects. Rather, these studies provide clinical context for representing antithrombotic exposure as a time-dependent trade-off between bleeding liability and ischemic protection.

The sensitivity analysis further showed that the location of the switching boundary depends substantially on structural assumptions. Treatment-related bleeding amplification, baseline ischemic and bleeding hazard scales, and the relative weighting of bleeding and ischemic losses showed the strongest associations with the loss-minimizing switching time. This dependence is important because these quantities were not estimated from patient-level data in the present study. The simulated switching times should therefore be interpreted as conditional consequences of the reference parameterization rather than robust clinical thresholds. The sensitivity results also identify parameters that would require particular attention during future empirical calibration.

From a mathematical perspective, the model converts the competing bleeding and ischemic processes into an explicit time-dependent loss-minimization problem. Continuous vulnerability functions allow hemoglobin and renal function to influence the system without imposing abrupt mathematical thresholds at clinically recognized risk cutoffs, while latent states retain information from preceding hazard exposure. The resulting zero level set of the expected-loss difference provides a theoretical switching boundary that can move over time even when observable covariates remain unchanged. This formulation provides a tractable structure for subsequent estimation and testing with longitudinal PCI data.

The present analysis should therefore be viewed as a structural demonstration rather than evidence for a particular de-escalation schedule. The numerical experiments establish that the proposed system can generate internally consistent, time-varying switching behavior and that this behavior remains numerically stable under the evaluated solver settings. They do not establish calibration, discrimination, causal treatment effects, or clinical benefit. A subsequent patient-level study would be required to estimate the structural parameters, evaluate whether the proposed dynamical formulation adds information beyond established risk models, and determine whether its switching boundary has clinically meaningful decision utility.

4.1 Strengths

The framework uses a parsimonious set of clinically interpretable variables supported by established post-PCI risk literature and places the switching analysis after the 30-day post-ACS landmark. It explicitly distinguishes literature-supported clinical anchors from theoretical structural parameters and evaluates the sensitivity of the loss-minimizing switching time to assumptions about hazards, treatment effects, and relative loss weights. Continuous vulnerability functions avoid imposing clinical risk thresholds as abrupt mathematical decision rules, while the dynamical formulation allows the switching boundary to evolve with elapsed time. The computational implementation is reproducible and includes prespecified grids, illustrative scenarios, structural sensitivity analysis, state perturbations, and numerical solver checks, providing a transparent framework that can subsequently be estimated and tested using patient-level PCI data.

4.2 Limitations

Several limitations should be considered. First, the model was not fitted to patient-level PCI outcomes; consequently, the reported probabilities, latent-state trajectories, and switching times are conditional simulation outputs rather than calibrated clinical estimates. Second, several structural quantities, including baseline hazard scales, treatment-effect coefficients, temporal decay parameters, latent-state coefficients, relative loss weights, and switching costs were specified theoretically and require empirical estimation. Third, treatment intensity was represented as a normalized exposure state rather than a drug- or dose-specific intervention, and the de-escalated state should therefore not be interpreted as a specific pharmacological conversion. Fourth, the latent bleeding and ischemic states are dimensionless mathematical constructs and have not been validated as measurable biological states. Fifth, hemoglobin and eGFR were treated as observable model inputs without jointly modelling their serial clinical evolution, while prior bleeding was represented as a simplified binary state. Important determinants and competing events including age, sex, diabetes, platelet count, procedural complexity, oral anticoagulation, recurrent revascularization, genotype, and death were not incorporated into the core proof-of-concept. Finally, published trials differ substantially in populations, treatment strategies, timing, and endpoint definitions and therefore cannot be treated as exchangeable calibration data. Clinical evaluation will require patient-level parameter estimation, explicit handling of time-dependent treatment exposure and confounding by indication, comparison with established risk models, external validation, and prospective assessment of decision utility.

5. Conclusions

This study demonstrates the feasibility of representing the post-PCI bleeding–ischemia trade-off as a time-dependent switching problem. Under the reference parameterization, lower hemoglobin, impaired renal function, and previous bleeding shifted the model-implied loss-minimizing switching time earlier, whereas lower-vulnerability profiles permitted later switching as the ischemic component declined over time. The resulting switching boundary provides a theoretical representation of how clinical vulnerability, elapsed post-PCI time, treatment exposure, and explicit loss assumptions may jointly influence a switching decision. These findings are conditional simulation outputs rather than estimates of clinically optimal treatment timing or individual patient risk. Patient-level parameter estimation, comparison with established risk models, clinical calibration, and external validation are required before the framework can be evaluated for predictive or decision-support use.

Data, Code, and Reproducibility

All simulation-derived numerical results reported in this study can be reproduced using the accompanying Python code. The repository includes the code required to generate the hemoglobin–eGFR switching surface, illustrative scenario analyses, Latin-hypercube parameter sampling, global and local sensitivity analyses, state perturbations, numerical solver checks, figures, and supplementary synthetic model-recovery analysis. Parameter definitions, reference values, and model equations are documented to support reproducibility. No patient-level data were used in the present analysis, and no patient-level data are distributed with the repository.

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Supplementary Table S1. Model Parameters and Evidence Status

Parameter	Unit	Meaning	Reference value or anchor	Evidence status
Hb	g/dL	Observed clinical variable	ARC-HBR Hb <11 g/dL major HBR criterion	Literature-derived clinical anchor
eGFR	mL/min/1.73m ²	Observed clinical variable	ARC-HBR <30 major; 30–59 minor HBR	Literature-derived clinical anchor
P	0/1	Prior spontaneous bleeding history	ARC-HBR/PRECISE-DAPT	Literature-derived variable definition
t0	days	Earliest de-escalation landmark	30 days	ESC 2023 boundary
b0	annualized hazard scale	Baseline bleeding hazard scale	0.026	Provisional structural coefficient calibrated only to keep simulations within published trial-scale ranges
i0	annualized hazard scale	Baseline ischemic hazard scale	0.090	Provisional structural coefficient
etaB	dimensionless	Potent-therapy bleeding amplification	0.70	Provisional structural coefficient
etaI	dimensionless	Potent-therapy ischemic protection	1.15	Provisional structural coefficient
deltaI	per month	Post-landmark ischemic hazard decay	0.180	Provisional structural coefficient
lambdaB	utility weight	Bleeding loss weight	1.00	Illustrative decision weight
lambdaI	utility weight	Ischemic loss weight	1.70	Illustrative decision weight
cs	utility	Switching penalty	0.003	Illustrative decision cost

Supplementary Table S2. Literature and CSV Provenance

Year	Used for	Anchor or role
2023	Clinical timing and de-escalation logic	De-escalation may be considered beyond 30 days in selected ACS patients to reduce bleeding; first-30-day de-escalation is not recommended.
2019	Hemoglobin, eGFR, prior bleeding thresholds	Hb <11 g/dL and eGFR <30 mL/min are major HBR criteria; eGFR 30–59 and mild anemia are minor criteria; prior spontaneous bleeding is a criterion.
2017	Variable selection	Age, hemoglobin, creatinine clearance, white-cell count, and previous spontaneous bleeding form the five-item PRECISE-DAPT score.
2022	External bleeding-risk context	PRECISE-DAPT remained informative under DAPT and ticagrelor-monotherapy strategies.
2016	Dual-risk renal signal	Renal dysfunction predicted both coronary thrombotic events and major bleeding; anemia predicted major bleeding.
2016	Benefit–risk decision precedent	Extended thienopyridine reduced ischemia but increased bleeding; the study formalized a benefit–risk difference after PCI.
2017	De-escalation event-rate anchor	At 1 year, the composite endpoint was 13.4% vs 26.3%; BARC ≥ 2 bleeding 4.0% vs 14.9% after switching vs unchanged therapy.
2017	Guided de-escalation precedent	Platelet-function-guided prasugrel-to-clopidogrel de-escalation was non-inferior for net clinical benefit.
2019	Bleeding/ischemia anchor	After 3 months of DAPT, BARC 2/3/5 bleeding was 4.0% with ticagrelor monotherapy vs 7.1% with ticagrelor plus aspirin; death/MI/stroke was 3.9% in both groups.
2019	Bleeding/ischemia anchor	At 12 months, MACCE was 2.9% vs 2.5% and bleeding 2.0% vs 3.4% for P2Y12 monotherapy vs DAPT.
2019	Short-DAPT anchor	One-month DAPT followed by clopidogrel monotherapy produced a 2.4% vs 3.7% composite cardiovascular/bleeding endpoint at 1 year.
2020	Dose de-escalation anchor	Prasugrel dose de-escalation reduced net adverse clinical events (7.2% vs 10.1%) mainly through less bleeding without increased ischemic risk.
2021	Primary literature event-rate anchor	From month 1 to 12, net events were 4.6% vs 8.2%, ischemic composite 2.1% vs 3.1%, and BARC 2/3/5 bleeding 3.0% vs 5.6%.
2021	HBR abbreviated-DAPT anchor	Major or clinically relevant nonmajor bleeding was 6.5% vs 9.4% with abbreviated vs standard therapy, while major cardiac/cerebral outcomes were non-inferior.
2011	Outcome definition	Standardized Bleeding Academic Research Consortium definitions for cardiovascular trials.
2016	Hemoglobin biological rationale	Across 17 studies and 68,528 patients, anemia was associated with increased long-term ischemic and bleeding events.
2019	Hemoglobin and treatment-disruption rationale	Baseline anemia was associated with higher adjusted MACE and major bleeding after PCI.

Supplementary Table S3. Global Sensitivity

parameter	PRCC_optimal_switch_month	p_value
treatment_bleed	-0.8956	0.0
i0	0.8419	0.0
b0	-0.7781	0.0
lambda_bleed	-0.7305	0.0
lambda_isch	0.6948	0.0
w_g_I	0.5619	0.0
w_g_B	-0.5157	0.0
ischemic_decay	-0.4994	0.0
w_hb_B	-0.4535	0.0
w_hb_I	0.3776	0.0
treatment_ischemia	0.1665	0.0028
rB	-0.0591	0.2915
w_pb_B	-0.0481	0.3911
kI	-0.0446	0.4265
kB	-0.0439	0.4334

Supplementary Table S4. Synthetic AIC

model	AIC_synthetic	RSS	fitted_parameters	delta_AIC
Shared exponential ischemic-decay model	-284.3416	0.0001	3	0.0
Full competing-hazard structural model	-282.3416	0.0001	4	2.0
Independent linear hazard model	-272.8696	0.0002	4	11.4721
Constant competing-hazard model	-222.5897	0.0019	2	61.7519